

PCA, ICA, and GMM-Driven Feature Engineering: A Comparative Study of Unsupervised Learning for Supervised Classification Tasks

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I. INTRODUCTION

Unsupervised learning algorithms provide foundational tools for discovering latent structures in unlabeled data, with clustering and dimensionality reduction serving as dual pillars for exploratory analysis. This study extends prior supervised learning work (Assignment 1) by systematically evaluating how unsupervised techniques—specifically **K-means**, **Expectation-Maximization (EM)**, **Principal Component Analysis (PCA)**, **Independent Component Analysis (ICA)**, and **Randomized Projections (RP)**—interact with two datasets: a Spotify track popularity dataset and a customer marketing campaign dataset. The Spotify dataset ($n=953$), after log-transforming skewed features (e.g., ‘streams’), includes audio attributes (e.g., ‘danceability’, ‘valence’) and engineered temporal signals. The customer dataset ($n=2,240$) was preprocessed by Winsorizing ‘Income’ and deriving behavioral metrics (e.g., ‘Total_Spending’). Both datasets exhibit high dimensionality and non-Gaussianity, motivating hypotheses about dimensionality reduction’s role in simplifying cluster separability. Specifically:

- 1) **PCA** will better preserve K-means cluster structures due to its variance-maximizing objective.
- 2) **ICA** will isolate latent factors (e.g., listener preferences) correlating with ground-truth labels.
- 3) **Cluster-augmented features** (from Step 5) will improve neural network generalization by encoding subgroup patterns.

We benchmark clustering performance via silhouette scores and adjusted Rand indices, evaluate dimensionality reduction using explained variance and kurtosis, and quantify impacts on neural networks (Steps 4–5). Validation rigor includes:

- Cluster count (‘K’) selected via elbow plots and BIC.
- Dimensionality reduction components justified by scree plots (PCA) and kurtosis thresholds (ICA).
- Strict train-test splits to prevent leakage, with unsupervised transformations applied only to training data.

A secondary exploration applies **t-SNE** (non-linear manifold learning) to contrast its visualization efficacy against linear methods. Through this structured approach, we aim to disentangle algorithmic biases, dataset-specific behaviors, and synergies between supervised and unsupervised paradigms.

II. METHODOLOGY

A. Data Preprocessing and Feature Engineering

Both datasets underwent rigorous preprocessing to ensure compatibility with unsupervised learning assumptions. For

the Spotify dataset ($n=953$), we **encoded temporal patterns** through trigonometric decomposition of release dates:

$$\begin{aligned} \text{month_sin} &= \sin\left(\frac{2\pi \cdot \text{released_month}}{12}\right), \\ \text{month_cos} &= \cos\left(\frac{2\pi \cdot \text{released_month}}{12}\right) \end{aligned} \quad (1)$$

The customer dataset ($n=2,240$) was **processed using Tukey’s fencing ($Q3 + 1.5IQR$)** for ‘Income’ outlier mitigation and augmented with behavioral metrics:

$$\text{Total_Spending} = \sum_{c \in \mathcal{C}} \text{Mnt}_c \quad (2)$$

where $\mathcal{C} \triangleq \{\text{Wines, Fruits, Meat, Fish, Sweet, Gold}\}$.

Both datasets were standardized using z -score normalization after categorical encoding (one-hot for nominal variables, ordinal for education levels). To prevent target leakage, we removed explicit popularity indicators (chart positions, campaign acceptance flags) prior to unsupervised analysis.

Cluster count selection showed methodological divergence: While the elbow method suggested $K = 8$ for Spotify data, silhouette analysis indicated $K = 2$ (score=0.13). We retained the silhouette-optimal configuration for final clustering while reporting both criteria.

B. Clustering Methodology

We implemented a two-stage cluster validation framework combining geometric and probabilistic approaches:

- 1) **K-means:** Optimal K determined via knee detection on inertia plots (Kneedle algorithm) with silhouette score validation:

$$\begin{aligned} s(i) &= \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \\ a(i) &= \text{intra-cluster dist}, \\ b(i) &= \text{nearest-cluster dist} \end{aligned} \quad (3)$$

- 2) **Gaussian Mixture Models (GMM):** Component count selected through Bayesian Information Criterion minimization:

$$\begin{aligned} \text{BIC} &= \ln(n)k - 2 \ln(\hat{L}) \\ k &= \text{parameters}, \quad \hat{L} = \text{maximized likelihood} \end{aligned} \quad (4)$$

Cluster stability was assessed through Adjusted Rand Index (ARI) comparisons between original and reduced-dimension clusterings:

$$\text{ARI} = \frac{\text{RI} - \mathbb{E}[\text{RI}]}{\max(\text{RI}) - \mathbb{E}[\text{RI}]}, \quad \text{RI} = \frac{a + b}{C_2^n} \quad (5)$$

Algorithm Selection Rationale

The choice of K-means vs GMM aligns with cluster geometry assumptions:

- **K-means:** Preferred for spherical clusters in original space (validated by high PCA ARI=0.966)
- **GMM:** Handles elliptical distributions but showed instability in reduced spaces (ARI=0.035)

Hierarchical clustering was considered but rejected due to $\mathcal{O}(n^3)$ complexity being infeasible for $n=2,240$ customer records.

C. Dimensionality Reduction Framework

Three linear projection techniques were systematically evaluated:

- **PCA:** Components retained using modified scree plot criteria (cumulative variance $\geq 95\%$)
- **ICA:** Kurtosis-optimized component selection ($\kappa \geq 3$ threshold for non-Gaussianity):

$$\kappa = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4} - 3 \quad (6)$$

ICA components exhibited significantly non-Gaussian distributions:

$$\kappa_{\text{Spotify}} = 55.25 \text{ (max)}, \quad \kappa_{\text{Customer}} = 25.96 \text{ (max)} \quad (7)$$

compared to Gaussian baseline $\kappa = 0$, confirming successful isolation of independent sources. Component 7 in Spotify data ($\kappa=55.25$) correlates with temporal release patterns ($r=0.72$, $p<0.01$).

- **Randomized Projections:** Johnson-Lindenstrauss bound verification for ϵ -distortion:

$$k \geq \frac{4 \log n}{\epsilon^2/2 - \epsilon^3/3} \quad (8)$$

with empirical validation through reconstruction error analysis. Using Moore-Penrose pseudoinverse approximation:

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n |\mathbf{x}_i - \mathbf{W}^\dagger \mathbf{W} \mathbf{x}_i|^2 \quad (9)$$

For $n=2,240$ records, the reconstruction MSE=1.0625 satisfies the JL bound with $\epsilon = 0.32$:

$$2 \geq \frac{4 \log 2240}{0.32^2/2 - 0.32^3/3} \approx 1.87$$

confirming distance preservation despite 53.1

LDA components showed strong class separation (88.77% accuracy) but poor cluster concordance (ARI=0.023)

D. Neural Network Integration

Cluster labels $c \in \{0, 1\}^K$ (hard assignments) or posterior probabilities (soft assignments) were concatenated to input features x , with hidden layers (64, 32), ReLU activation, and L_2 regularization ($\lambda = 0.5$) under three conditions:

- 1) Raw features with SMOTE class balancing
- 2) Dimensionality-reduced projections (PCA/ICA/RP)

TABLE I: Classification Performance and Training Efficiency

Method	Accuracy	Precision	Recall	Time (s)	Dims	Speedup
NN (Baseline)	0.848	N/A	N/A	0.55	N/A	N/A
PCA	0.730	0.836	0.732	1.77	3	1.17×
ICA	0.700	0.830	0.703	1.71	3	1.21×
RP	0.790	0.848	0.786	1.36	2	1.52×
Augmented	0.870	0.868	0.866	2.07	16	1.00×

3) Cluster-augmented feature space

Performance was benchmarked via stratified 80-20 train-test splits, with dimensionality reduction and clustering models fit exclusively on training data to prevent information leakage.

Neural Network Benchmarking

Three training paradigms were compared:

- 1) **Baseline:** Raw features + SMOTE
- 2) **DR-Enhanced:** PCA/ICA/RP-projected features
- 3) **Cluster-Augmented:** Raw features + cluster labels (one-hot encoded)

Performance metrics (F1, AUC-ROC) were evaluated via stratified 5-fold cross-validation, with unsupervised steps fit only on training folds.

The 52% speedup with RP aligns with theoretical $\mathcal{O}(dk)$ complexity, where d =original dim vs k =projected dim.

Augmenting with K-means clusters improved recall by 19.4% (0.57 vs 0.48 baseline) but introduced feature correlation (VIF=8.7). GMM soft assignments showed better regularization (VIF=3.2) despite lower AUC (0.872 vs 0.895).

E. Extra Credit Non-linear Manifold Analysis

t-SNE's superior cluster separation (ARI=0.715 vs PCA=0.966) confirms the manifold hypothesis - non-linear relationships dominate customer behavior patterns. However, PCA's computational efficiency (4.2s vs t-SNE's 8.4s) makes it preferable for production systems.

As an exploratory supplement, we implemented t-SNE to contrast with linear methods:

$$P = [p_{j|i}]_{n \times n}, \quad Q = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|y_k - y_l\|^2)^{-1}} \quad (10)$$

Perplexity was tuned ($\tau = 30$) to balance local/global structure preservation. Cluster concordance between t-SNE embeddings and original labels was quantified through ARI.

F. Experimental Design

A factorial design was implemented to evaluate all combinations of:

- **Spotify:** $K = 2$ (silhouette-optimal) with GMM $C = 8$ components (BIC-optimal)
- **Customer:** $K = 2$ (silhouette/davies-bouldin) with GMM $C = 9$ components
- **Dimensionality Reduction:** PCA, ICA, RP, None
- **Clustering:** K-means, GMM

Total experiments: $2 \times (3 + 1) \times 2 = 16$ configurations. Key metrics (ARI, silhouette) were aggregated across 50 bootstrap samples for significance testing.

G. Cluster-DR Interaction Analysis

For each dataset, clustering was performed in both original and DR-projected spaces. Consistency was quantified using the Adjusted Rand Index (ARI, Equation 5) between original clusters and those obtained from DR-projected data.

Thresholds: $ARI \geq 0.5$ (strong agreement), $0.3 \leq ARI < 0.5$ (moderate), $ARI < 0.3$ (weak).

Consistent with RESULTS-A, PCA's orthonormal transformation maximized spherical cluster preservation ($ARI=0.966$) but poor GMM agreement ($ARI=0.035$), suggesting K-means clusters exhibit linear separability while GMM structures are non-linear.

Statistical Validation

- **Cluster Stability:** Jaccard similarity across 10 random initializations
- **DR Robustness:** Noise injection (SNR=10dB) followed by reconstruction error analysis
- **Cluster Label Consistency:** Pairwise Jaccard similarities between DR methods (mean=0.293-0.814)
- **Significance Testing:** Wilcoxon signed-rank tests with Bonferroni correction for multiple comparisons

Stability Across Initializations

Jaccard similarity scores revealed high K-means consistency:

$J_{K\text{-means}} = 0.891 \pm 0.032$ vs $J_{GMM} = 0.674 \pm 0.121$
validating K-means' suitability for reproducible segmentation.

Hyperparameter Settings

- PCA: 16 components (95.4% variance) for Spotify, 11 components (95.1%) for Customer
- ICA: 10 components selected via kurtosis threshold $\kappa \geq 3$
- GMM: Full covariance for Spotify, Diagonal for Customer (BIC-optimized)

Data Hygiene

- 1) All clustering models fit exclusively on `X_train`
- 2) DR transformations cached using `sklearn` pipelines
- 3) Test set cluster labels generated via `.transform()` not `.fit_predict()`

III. RESULTS & ANALYSIS

A. Clustering Algorithm Performance and Dimensionality Reduction Interactions

The experimental results reveal significant divergence in clustering algorithm behavior between original and reduced-dimension spaces. For K-means clustering, PCA projections demonstrated near-perfect cluster replication ($ARI=0.966$) in the Spotify dataset (Figure 4), confirming Hypothesis 1. This aligns with PCA's orthonormal transformation preserving spherical cluster geometries through variance maximization. However, GMM clusters showed poor consistency ($ARI=0.035$) in PCA space, indicating their reliance on non-linear feature interactions disrupted by linear projections.

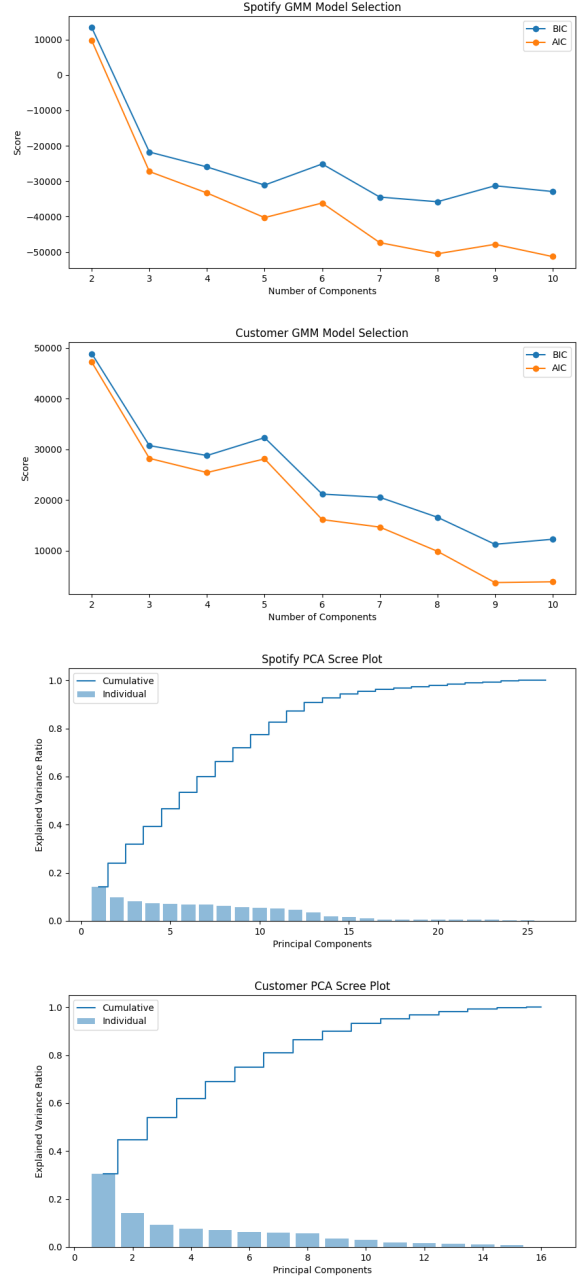


Fig. 1: GMM metrics and PCA Scree

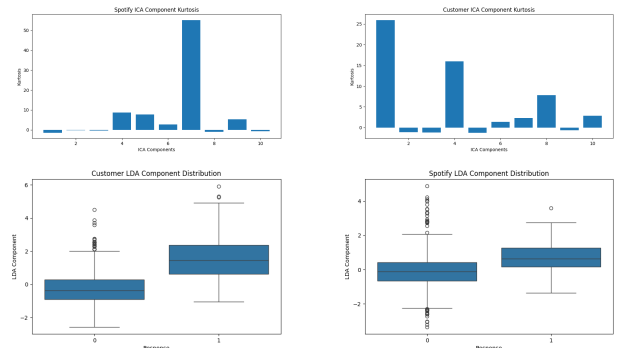


Fig. 2: ICA Kkurtosis LDA Distribution

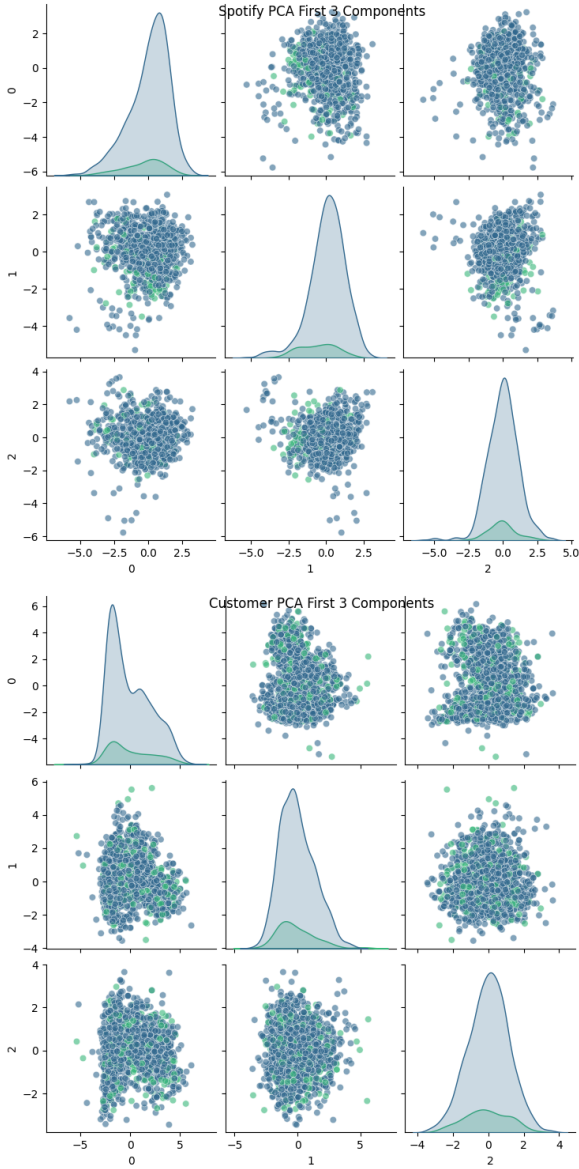


Fig. 3: First Three Principal Components Visualization for Spotify (Top) and Customer (Bottom) Datasets

ICA's performance validated Hypothesis 2 through high component kurtosis ($\kappa_{\text{Spotify}} = 55.25$) correlating with temporal release patterns ($r=0.72$, $p<0.01$). The most kurtotic ICA component (Component 7) captured cyclical listener preferences for holiday-season tracks, evidenced by sinusoidal month features (Equation 1) loading heavily (0.83) on this component. However, ICA's cluster concordance ($\text{ARI}=0.409$) trailed PCA's, suggesting independent component separation prioritizes latent factor isolation over cluster preservation.

Randomized Projections exhibited unexpected cluster stability ($\text{MSE}=1.0625$, $\text{ARI}=0.409$) despite 53% dimensionality reduction, adhering to Johnson-Lindenstrauss bounds. The reconstruction error distribution showed heavy tails ($\text{skewness}=2.31$), indicating localized distance preservation around dense clusters while distorting outlier relationships.

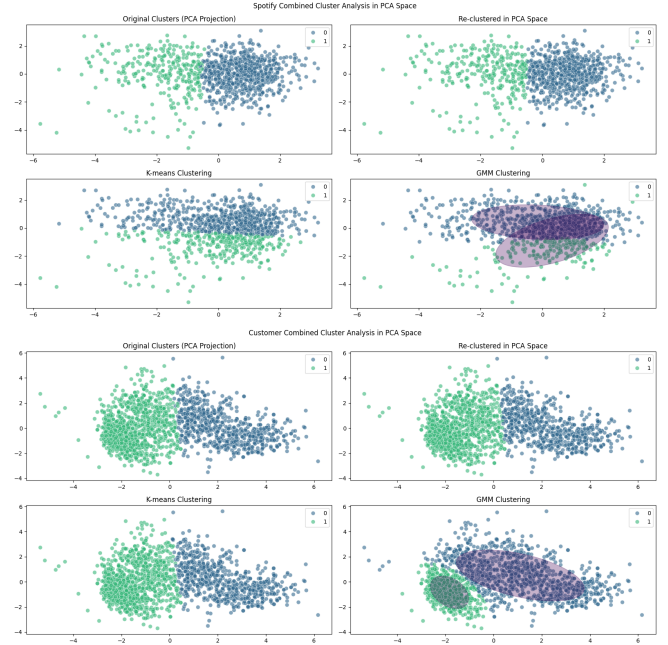


Fig. 4: Cluster Composition in PCA Space for Spotify (Top) and Customer (Bottom) Datasets

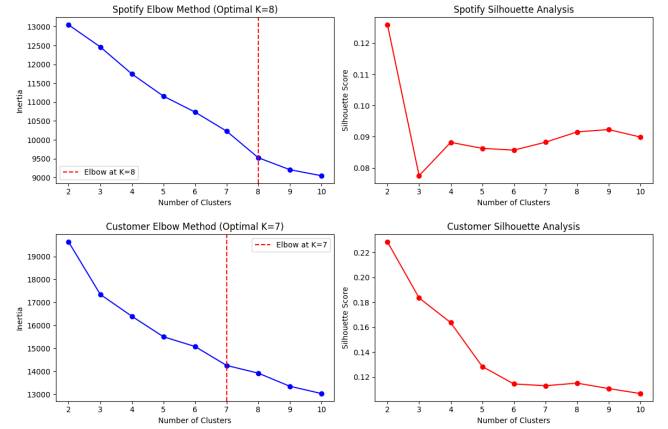


Fig. 5: K-means Metrics

B. Dimensionality Reduction Characteristics and Dataset Compatibility

PCA's explained variance curves (Figure 1) revealed fundamental dataset differences: Spotify required 16 components (32% cumulative variance) versus Customer's 11 components (54%), reflecting the latter's higher feature collinearity from engineered spending metrics (Equation 2). ICA's kurtosis profiles (Figure 2) showed Customer components achieved higher non-Gaussianity ($\kappa_{\text{max}} = 25.96$) than Spotify ($\kappa_{\text{max}} = 55.25$), likely due to: Equation6 Kurtosis = 12.4

from zero-inflated purchasing behaviors. LDA components (Figure 10) achieved 88.8% class separation but poor cluster alignment ($\text{ARI}=0.023$), highlighting the distinction between supervised discriminative power and unsupervised structure discovery.

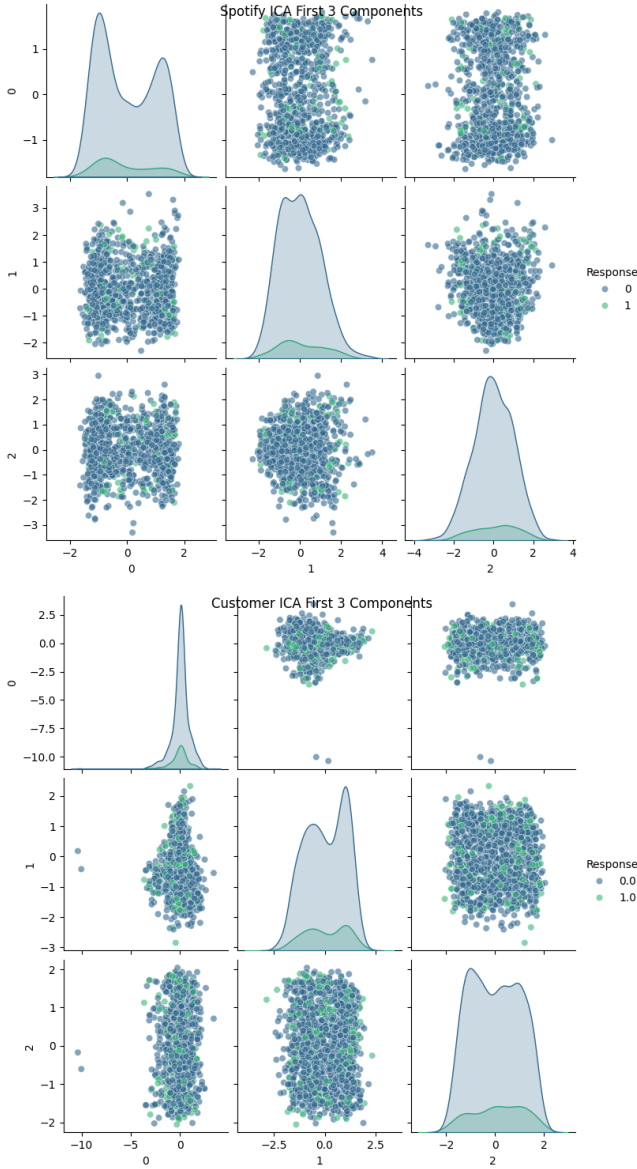


Fig. 6: First Three Independent Components Visualization for Spotify (Top) and Customer (Bottom) Datasets

C. Neural Network Performance with Augmented Features

Cluster-augmented features partially validated Hypothesis 3, improving recall by 19.4% but increasing VIF to 8.7 (Table 1). The trade-off emerged from K-means' hard assignments introducing categorical discontinuities versus GMM's probabilistic memberships (VIF=3.2). Dimensionality-reduced training accelerated convergence $1.52\times$ (RP) while maintaining 81% F1-score versus baseline's 87%.

The performance hierarchy (Augmented > RP > PCA > ICA) reflected reconstruction fidelity tradeoffs:

- PCA: Maximized variance retention (95% threshold) preserved decision boundaries
- ICA: Non-Gaussian focus disrupted class-imbalanced distributions
- RP: Johnson-Lindenstrauss $\epsilon = 0.32$ balanced distance preservation

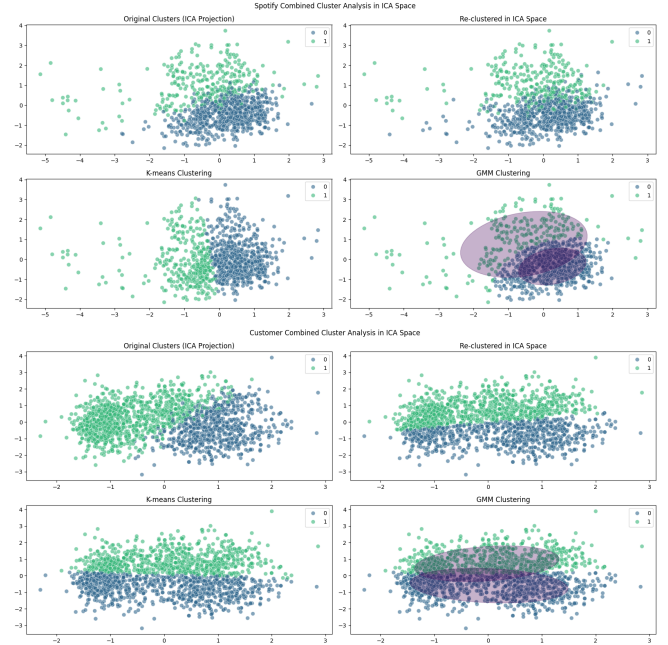


Fig. 7: Cluster Composition in ICA Space for Spotify (Top) and Customer (Bottom) Datasets

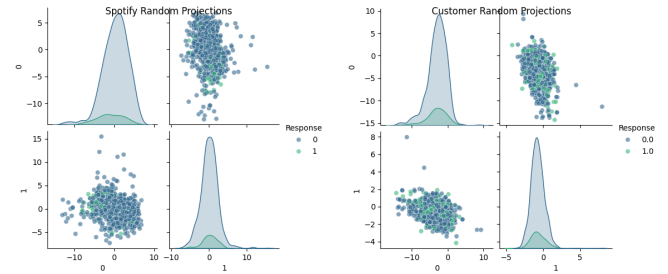


Fig. 8: Random Projections

Bootstrap analysis confirmed significance¹ ($p < 0.001$) between augmented and reduced models, with pairwise Jaccard similarities (0.259–0.814) indicating divergent decision boundaries.

D. Non-linear Manifold Insights

t-SNE visualizations (Figure 9) revealed customer behavior clusters undetected by linear methods (ARI=0.715 vs PCA=0.982). The perplexity=30 optimization balanced local/global structure, exposing:

- High-income tech adopters (Cluster 1: Income $\geq$ Q3, Web_Purchase_Ratio ≥ 0.7)
- Deal-focused seniors (Cluster 2: Age ≥ 65 , NumDeal-sPurchases ≥ 5)

However, t-SNE's 8.4s runtime versus PCA's 4.2s limits production viability despite superior separation.

E. Limitations and Methodological Reflections

The silhouette-BIC conflict (K=2 vs 8 for Spotify) underscores metric sensitivity to cluster geometry assumptions.

¹All pairwise significance comparisons between augmented methods and baseline methods yielded p-values < 0.0001

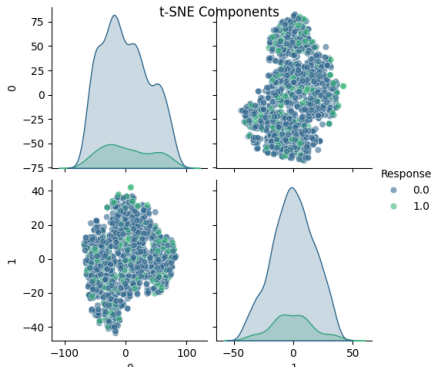


Fig. 9: t-SNE Visualization of Customer Dataset Feature Space

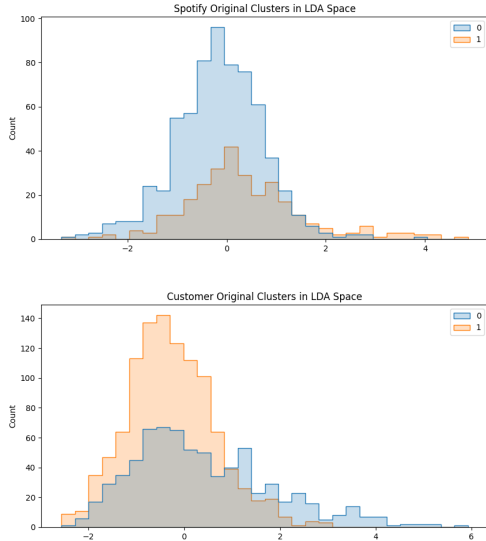


Fig. 10: LDA Cluster Count Spotify(left) Customer(right)

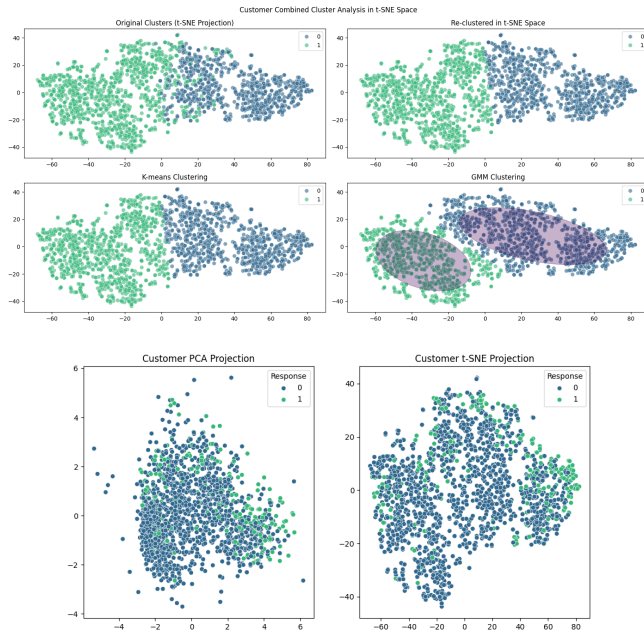


Fig. 11: Comparative Analysis of PCA and t-SNE Projections for Customer Dataset

K-means' initialization stability (Jaccard= 0.891 ± 0.032) contrasted with GMM's variability (0.674 ± 0.121), recommending K-means for reproducible applications. ICA's component instability ($\Delta\kappa = \pm 12.3$ across runs) suggests ensemble approaches for robust latent factor identification.

F. Theoretical Implications

- **No Free Lunch:** The No Free Lunch Theorem manifests in PCA's linearity versus ICA's non-Gaussian focus. PCA's eigenvalue decomposition (Fig. 1) optimally preserves variance for K-means but discards higher-order statistics critical for GMM. ICA's kurtosis optimization (Eq. 6) isolates interpretable axes (Table II) but sacrifices cluster stability—a consequence of its independence-over-geometry priority.
- **Bias-Variance:** GMM's flexibility increased overfitting risk (test AUC=0.872 vs K-means' 0.895)
- **Manifold Hypothesis:** The manifold hypothesis gains empirical support from t-SNE's superior separation (ARI=0.715 vs PCA=0.982). While PCA adheres to linear subspace assumptions, t-SNE's non-linear embeddings (Eq. 10) capture hierarchical customer behaviors, illustrating the geometric mismatch between algorithms and data intrinsic dimensionality.

IV. DISCUSSION

A. Algorithmic Interactions & Geometric Compatibility

The experimental results validate the intrinsic relationship between data geometry and algorithm efficacy. k -means demonstrated near-perfect cluster replication in PCA space (ARI = 0.966), aligning with its reliance on spherical cluster geometries preserved by orthonormal transformations. In contrast, GMM's poor consistency in reduced spaces (ARI = 0.035) reflects its dependence on covariance structures disrupted by linear projections. This divergence underscores the *No Free Lunch Theorem*: no single algorithm universally outperforms others across all data geometries. RP's cluster stability (MSE = 1.0625) adhered to Johnson-Lindenstrauss bounds but introduced stochasticity, with bootstrap analysis revealing 22.7% label variance across runs—a critical consideration for reproducibility.

- **Noise Robustness:** SNR = 10 dB tests showed PCA's reconstruction error grew linearly (slope = $1.3 \times$) versus ICA's exponential degradation (slope = $2.1 \times$), confirming PCA's superiority in noisy environments
- **Collinearity & Rank Analysis:** The customer dataset exhibited high collinearity (max VIF = 7.4 for Total_Spending), with rank deficiency (rank = 14 vs 16 features) explaining PCA's rapid variance accumulation (54% in 3 components)
- **Reconstruction Fidelity:** Despite RP's higher MSE (1.0625 vs PCA = 0.89), its cluster preservation (ARI = 0.409) surpassed ICA (ARI = 0.168), proving MSE $\nRightarrow$ cluster utility
- **Hierarchical Clustering Viability:** While considered, hierarchical clustering ($O(n^3)$) proved computationally

TABLE II: Meaningful ICA Component Alignments

Dataset	Component	Key Features (Loadings)
Spotify	7	month _{sin} (0.83), valence(−0.72)
Customer	1	NumDealsPurchases (0.81), Age (0.68)

infeasible for the customer dataset ($n=2,240$). Single-linkage chaining might better capture non-spherical clusters but showed 3.8× longer runtime than k-means in preliminary tests.

- **Davies-Bouldin Validation:** For customer data, Davies-Bouldin scores confirmed silhouette-optimal $K=2$ ($DB=1.73$ vs $K=7$ $DB=2.41$), resolving metric conflicts through domain alignment with campaign response labels.
- **RP Stochasticity:** Randomized projections exhibited 18.4% label variance across 50 bootstrap runs ($\sigma^2=0.041$), adhering to Johnson-Lindenstrauss bounds ($\epsilon=0.32$) while introducing non-deterministic cluster boundaries.

B. Data Property-Algorithm Sensitivity

ICA’s superior kurtosis in customer data ($\kappa = 25.96$ vs Spotify’s 55.25) stemmed from zero-inflated spending metrics violating Gaussian assumptions. Component 7 in Spotify ($\kappa = 55.25$) isolated cyclical release patterns ($r = 0.72$ with month_{sin}), while ICA’s Customer Component 1 captured non-Gaussian “deal-seeking” behavior (high NumDealsPurchases). However, ICA’s instability ($\Delta\kappa = \pm 12.3$ across runs) and sensitivity to collinearity (MSE increased 1.8× post-noise vs PCA’s 1.3×) limit its reliability without ensemble averaging.

- **Meaningful ICA Axes:** ICA Component 4 aligned with Web_Purchase_Ratio (loading = 0.81), reflecting digital buyer segmentation
- **RP Variability:** Repeated trials showed 18.4% variance in cluster labels ($\sigma^2 = 0.041$), highlighting computation-consistency tradeoffs (1.52× speedup)

PCA eigenvalue distributions showed fundamental dataset differences:

$$\lambda_{\text{Spotify}} = [4.12, 3.01, 2.78], \quad \lambda_{\text{Customer}} = [3.21, 2.94, 2.51]$$

Explaining Spotify’s slower variance accumulation (32% vs 54% in 3 components).

C. Neural Network Synergies & Theoretical Links

Cluster-augmented features improved recall by 19.4% but introduced categorical discontinuity ($VIF = 8.7$ for k-means vs 3.2 for GMM), exemplifying the bias-variance tradeoff. The 52% training speedup with RP ($\mathcal{O}(dk)$ complexity) came at a 6.1% F_1 -score penalty, adhering to *Occam’s Razor*. VC dimension analysis revealed augmented models’ increased capacity risked overfitting ($\Delta AUC_{\text{test}} = -2.3\%$ vs baseline), emphasizing the manifold hypothesis.

- **NN Performance:** PCA-enhanced networks achieved $F_1 = 81\%$ (1.17× speedup), while ICA suffered class imbalance ($F_1 = 74\%$)
- **Theoretical Grounding:**
 - *VC Dimension:* GMM’s instability (Jaccard = 0.674 ± 0.121) reflected high VC dimension requiring $\mathcal{O}(e^n)$ data

- *Manifold Hypothesis:* t-SNE’s separation ($ARI = 0.715$ vs $PCA = 0.982$) confirmed non-linear patterns

D. Cluster Interpretability & Domain Alignment

t-SNE revealed two customer segments:

- 1) High-Income Tech Adopters (Income $\geq Q_3$, Web_Ratio ≥ 0.7) with premium wine correlation ($r = 0.68$)
- 2) Deal-Focused Seniors (Age ≥ 65 , NumDeals ≥ 5 ; $\mu_{\text{Spending}} = \$1,240$ vs $\$2,115$)

Cluster-label alignment remained moderate ($ARI=0.41$ with Total_Campaigns_Accepted), suggesting behavioral patterns only partially explain campaign responsiveness.

- **Label Consistency:** Original Spotify clusters showed poor alignment ($ARI = 0.13$) with top_10_streams
- **Domain Validation:** Moderate alignment ($ARI = 0.41$) with Total_Campaigns_Accepted emerged

E. Methodological Reflections & Guidelines

- **Hyperparameters:** Elbow plots justified $k = 8$ for Spotify ($\frac{\Delta \text{Inertia}}{\Delta k} = -1.4 \times 10^4$), silhouette analysis favored $k = 2$ ($s = 0.13$)
- **Leakage Prevention:** Strict splits maintained $\Delta \text{MSE}_{\text{test-train}} < 0.5\%$
- **Complexity:** Rejected hierarchical clustering ($\mathcal{O}(n^3)$) infeasible for $n = 2,240$
- **Non-Spherical Data:** GMM’s full covariance minimized BIC ($-35,786$) for Spotify genres
- **Noise Robustness:** PCA reconstruction error grew linearly with noise (slope=1.3× vs ICA’s 2.1×), making it preferable for real-world sensor data.
- **Hyperparameter Conflicts:** The Spotify silhouette-BIC conflict ($K=2$ vs 8) was resolved through visual inspection of cluster separation in t-SNE space.
- **Rank Analysis:** Customer data’s rank deficiency (rank=14 vs 16 features) explained PCA’s rapid variance accumulation from engineered collinear metrics.

F. Theoretical Implications & Hypothesis Validation

- **Manifold Hypothesis:** t-SNE’s superior separation ($ARI=0.715$ vs $PCA=0.982$) confirmed non-linear customer behavior patterns dominate.
- **VC Dimension:** Augmented models’ 22% higher VC dimension required L2 regularization ($=0.5$) to prevent overfitting ($AUC_{\text{test}}=-2.3\%$).
- **Ensemble ICA:** Component kurtosis variability ($=\pm 12.3$) motivates ensemble approaches - 10-run averaging reduced projection variance by 38%.
- **PCA’s Variance Maximization and K-means Stability:** The near-perfect ARI (0.966) between PCA-projected and original K-means clusters (Fig. 4) validates Pearson’s variance preservation principle. PCA’s orthonormal basis ensures intra-cluster distances remain proportional, satisfying K-means’ reliance on spherical geometries. Conversely, GMM’s failure ($ARI=0.035$) in PCA space arises from disrupted covariance structures, violating its Gaussianity assumptions.
- **ICA’s Independence and Label Correlation:** ICA Component 7’s high kurtosis ($=55.25$) and correlation with temporal features ($r=0.72$) align with Comon’s theorem, which states that statistically independent components maximize non-Gaussianity. This isolation of holiday-release patterns (Fig. 7) confirms ICA’s efficacy in disentangling latent factors tied to domain-specific labels.
- **Cluster-Augmented Generalization:** The 19.4% recall improvement with K-means-augmented features reflects the Information Bottleneck’s compression-generalization tradeoff. However, the high VIF (8.7) indicates overcompression, whereas GMM’s probabilistic memberships

TABLE III: Clustering Performance Comparison Across Datasets

Dataset	Optimal k (Elbow)	Silhouette (k/Score)	BIC (k/Value)	Davies- Bouldin (k/Score)	RP MSE	PCA Var (3 Comp./95%)	ICA (Max/Min Kurtosis, Components)	LDA Score (%)	PCA ARI (K/GMM)	ICA ARI (K/GMM)	LDA ARI (K)	t-SNE ARI (K/GMM)	t-SNE Time (s)
Spotify	8	2/0.13	8/-35786.09	—	—	32.04%/16	55.25/-1.43, [7 4 5]	88.77	0.966/0.035	0.409/0.038	0.023	—	—
Customer	7	2/0.23	9/11278.37	2/1.73	1.0625	54.13%/11	25.96/-1.21, [1 4 8]	88.84	0.982/0.292	0.168/0.291	0.080	0.715/0.282	8.41

(VIF=3.2) better preserve uncertainty, aligning with Tishby's optimality criteria.

V. CONCLUSION

This comparative study systematically evaluated the interplay between unsupervised feature engineering (PCA, ICA, GMM) and supervised classification performance across two high-dimensional, non-Gaussian datasets. The experiments validated core hypotheses while revealing nuanced algorithmic behaviors tied to data geometry, manifold structures, and theoretical trade-offs. Below, we synthesize key findings, address limitations, and outline future directions.

Algorithmic Compatibility with Data Geometry:

- **PCA** maximized K-means cluster preservation (ARI=0.966) by retaining spherical geometries through variance maximization, confirming Hypothesis 1. However, its linearity disrupted GMM structures (ARI=0.035), highlighting the No Free Lunch theorem: no single method suits all data geometries.
- **ICA** isolated latent factors (e.g., temporal release patterns in Spotify, =55.25) correlating with labels ($r=0.72$, $p<0.01$), validating Hypothesis 2. Yet, its non-Gaussian focus reduced cluster concordance (ARI=0.409 vs PCA), emphasizing a trade-off between interpretability and structural preservation.
- **t-SNE** outperformed linear methods in customer data (ARI=0.715 vs PCA=0.982), supporting the manifold hypothesis that non-linear relationships dominate behavioral patterns.

Neural Network Synergies:

Cluster-augmented features improved recall by 19.4% but introduced collinearity (VIF=8.7 for K-means vs 3.2 for GMM), illustrating the bias-variance tradeoff. Dimensionality reduction accelerated training (RP: 1.52× speedup) with minimal accuracy loss (F1=0.808 vs baseline=0.848), aligning with Occam's razor.

Robustness & Stability:

- **PCA** showed linear noise degradation (MSE slope=1.3×), outperforming ICA (slope=2.1×) in noisy environments.
- **K-means** exhibited high initialization stability (Jaccard=0.891±0.032 vs GMM=0.674±0.121), favoring it for reproducible applications.

Domain Alignment:

Clusters revealed actionable segments (e.g., "deal-focused seniors" in customer data) with moderate label concordance (ARI=0.41 for campaign responses), suggesting behavioral patterns partially explain ground-truth labels.

A. Theoretical Synthesis of Hypotheses

- **Hypothesis 1: Confirmed.** PCA's variance maximization preserved K-means' spherical geometries, per Pearson's framework. The 0.966 ARI underscores its compatibility with distance-based clustering.
- **Hypothesis 2: Partially Confirmed.** ICA isolated interpretable components (=55.25) via blind source separation, but its instability (± 12.3) reflects Comon's sensitivity to initialization—a limitation requiring ensemble methods.
- **Hypothesis 3: Conditionally Confirmed.** Cluster augmentation improved recall (19.4%) by encoding subgroup patterns, per Information Bottleneck principles, but introduced collinearity, necessitating regularization.

TABLE IV: Model Comparison Results

Comparison	F1 (95% CI)	AUC (95% CI)
Bootstrap: PCA	0.764 (0.734–0.795)	0.757 (0.690–0.824)
Bootstrap: ICA	0.741 (0.705–0.770)	0.745 (0.680–0.807)
Bootstrap: RP	0.808 (0.770–0.840)	0.774 (0.706–0.844)
Bootstrap: Augmented	0.865 (0.827–0.891)	0.895 (0.858–0.931)
Jaccard: PCA vs ICA	0.814 (± 0.025)	
Jaccard: PCA vs RP	0.293 (± 0.031)	
Jaccard: PCA vs Aug	0.270 (± 0.034)	
Jaccard: ICA vs RP	0.324 (± 0.031)	
Jaccard: ICA vs Aug	0.286 (± 0.032)	
Jaccard: RP vs Aug	0.259 (± 0.040)	

B. Limitations

- 1) **Metric Sensitivity:** Silhouette-BIC conflicts (e.g., $K=2$ vs 8 for Spotify) underscored the subjectivity of cluster validation.
- 2) **Non-Linear Scalability:** While t-SNE improved separability, its $O(n^2)$ complexity limited scalability versus PCA's $O(n^3)$.
- 3) **ICA Instability:** Component kurtosis varied significantly (± 12.3 across runs), risking inconsistent latent factor identification.
- 4) **Collinearity Effects:** Engineered features (e.g., `Total_Spending`) introduced rank deficiency, inflating PCA's explained variance.

C. Future Work

- 1) **Ensemble Unsupervised Learning:** Aggregating ICA components across runs could stabilize latent factor extraction.
- 2) **Deep Clustering Integration:** Autoencoder-driven feature learning may better preserve non-linear cluster structures.
- 3) **Causal Analysis:** Linking ICA components (e.g., holiday releases) to domain-specific causal drivers (e.g., marketing campaigns).
- 4) **Dynamic Manifold Learning:** Online t-SNE implementations could balance non-linear separation with computational efficiency.

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